Library Borrowing and Fine Reconciliation

1. Problem Statement

Develop a Python program to analyze library book borrowing records. The program should identify borrowed and returned books, calculate fines for overdue returns, and reconcile book availability. It should display the borrowing details, overdue days, fine amounts, and the final availability status of each book.

2. Objectives

1. To store library borrowing records using lists and tuples.
2. To identify books that have been returned and books that are still borrowed.
3. To calculate fines for overdue books.
4. To determine the availability of books in the library.
5. To display the borrowing records and total fine collected.

3. Assumptions

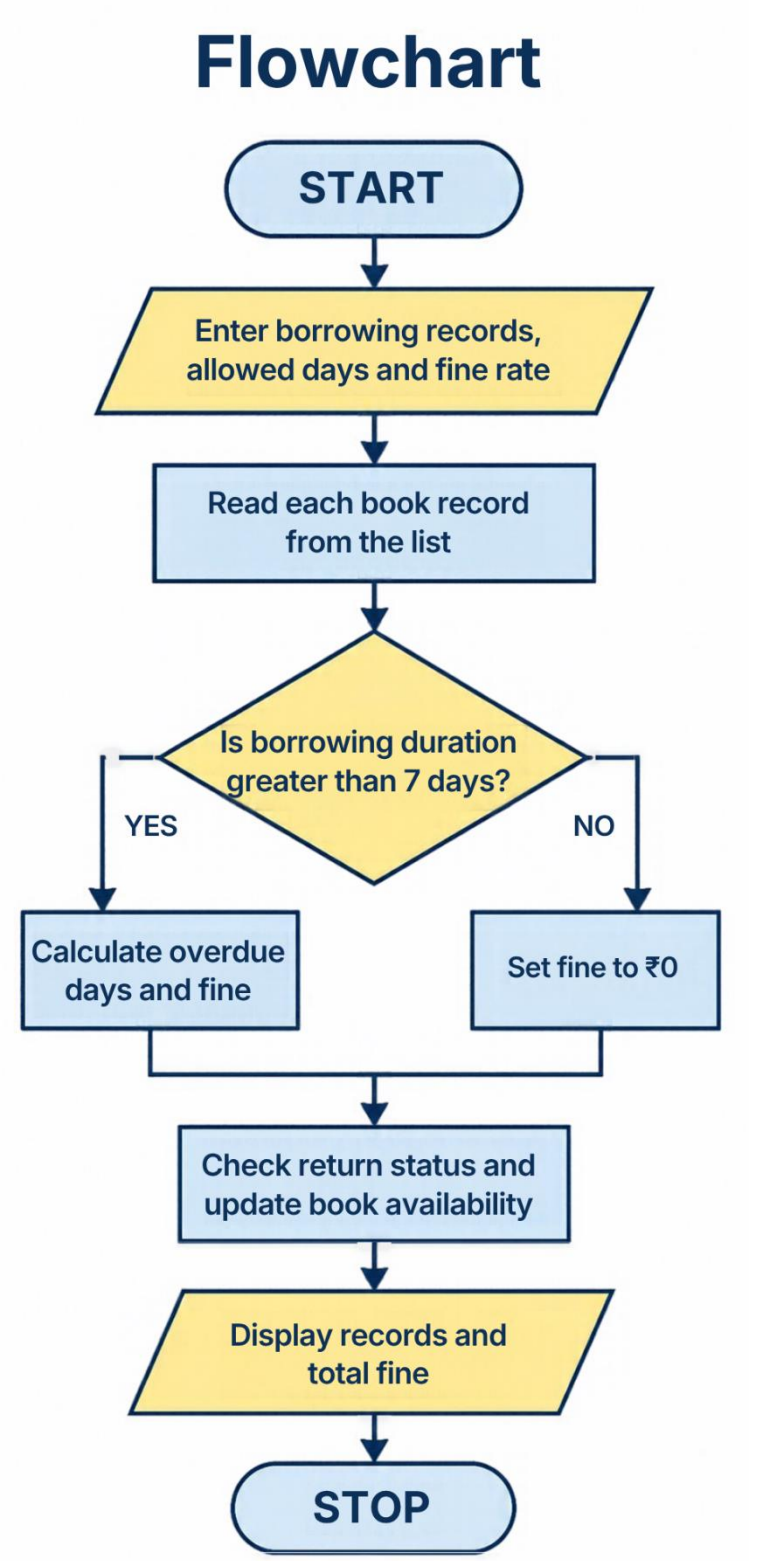
1. Each book has a unique book ID.
2. The allowed borrowing period is 7 days.
3. The fine is ₹5 per overdue day.
4. Returned books are available for borrowing again.
5. Books that have not been returned are marked as unavailable.
6. The borrowing duration is provided in days.
7. If a book is returned within 7 days, no fine is charged.

4. Algorithm

1. Start the program.
2. Store the borrowing records in a list of tuples.
3. Set the maximum borrowing period to 7 days.

4. Set the fine per overdue day to ₹5.
5. Initialize the total fine to zero.
6. Read each borrowing record using a loop.
7. Calculate overdue days by subtracting the allowed borrowing period from the borrowing duration.
8. If overdue days are greater than zero, calculate the fine.
9. Otherwise, set the fine to zero.
10. Check whether the book has been returned.
11. Display the book details, fine amount, and availability status.
12. Calculate and display the total fine.
13. Stop the program.

5. Flowchart



6. Pseudocode

START

```
SET allowed_days = 7
SET fine_per_day = 5
SET total_fine = 0

STORE borrowing records in a list

FOR each record in borrowing records:
    READ book ID, title, days borrowed, return status

    IF days borrowed > allowed_days:
        overdue_days = days borrowed - allowed_days
        fine = overdue_days * fine_per_day
    ELSE:
        overdue_days = 0
        fine = 0

    ADD fine to total_fine

    IF return status is "Yes":
        availability = "Available"
    ELSE:
        availability = "Not Available"

    DISPLAY book details, fine and availability
```

DISPLAY total_fine

STOP

7. Python Actual Code

```
# Library Borrowing and Fine Reconciliation

# Each tuple contains:
# (Book ID, Book Title, Days Borrowed, Returned)

records = [
    (101, "Python Basics", 10, "Yes"),
    (102, "Data Science", 5, "Yes"),
    (103, "Artificial Intelligence", 12, "No"),
    (104, "Web Development", 8, "Yes"),
    (105, "Machine Learning", 6, "No")
]

allowed_days = 7
fine_per_day = 5
total_fine = 0

print("LIBRARY BORROWING AND FINE REPORT")
```

```
print("-" * 65)

for record in records:
    book_id, title, days, returned = record

    overdue_days = max(0, days - allowed_days)
    fine = overdue_days * fine_per_day
    total_fine += fine

    if returned == "Yes":
        availability = "Available"
    else:
        availability = "Not Available"

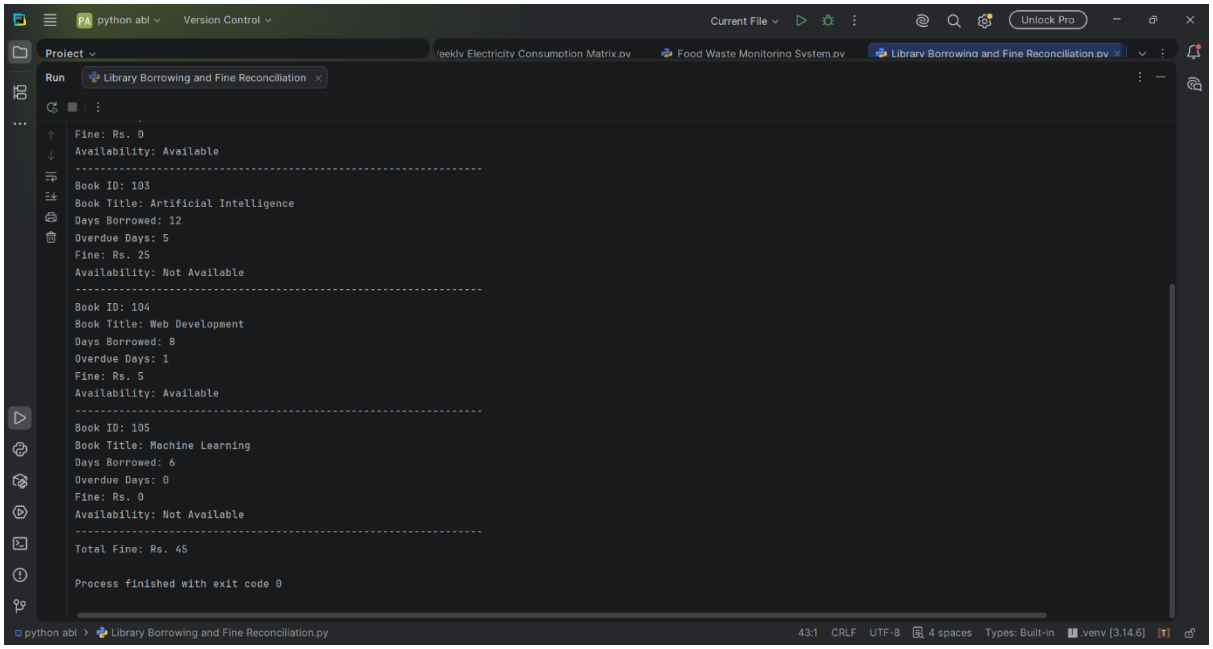
    print("Book ID:", book_id)
    print("Book Title:", title)
    print("Days Borrowed:", days)
    print("Overdue Days:", overdue_days)
    print("Fine: Rs.", fine)
    print("Availability:", availability)
    print("-" * 65)

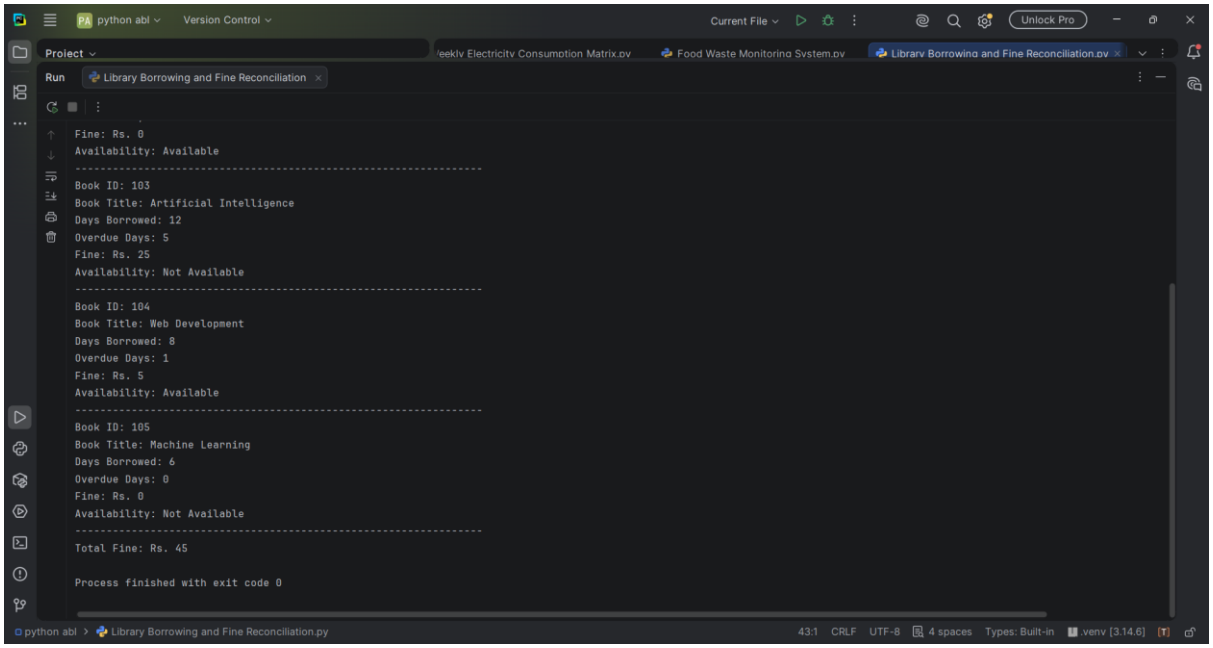
print("Total Fine: Rs.", total_fine)
```

8. Explanation of the Code

1. Creating records: The records list stores book details as tuples. Each tuple contains a book ID, title, borrowing duration, and return status.
2. Setting limits: allowed_days = 7 defines the borrowing period, and fine_per_day = 5 defines the fine rate.
3. Using a loop: The for loop processes each book record one by one.
4. Calculating overdue days: max(0, days - allowed_days) calculates overdue days without allowing a negative value.
5. Calculating the fine: The overdue days are multiplied by ₹5 to obtain the fine.
6. Updating the total: total_fine += fine adds each book's fine to the overall total.
7. Checking availability: The if-else condition marks returned books as available and unreturned books as not available.
8. Displaying results: The print() statements display each book's information, its fine, its availability, and the total fine.

9. Output





10. Conclusion

The Library Borrowing and Fine Reconciliation program successfully analyzes book borrowing records using Python. It calculates overdue days and fines, identifies returned and unreturned books, and displays their availability status. This program demonstrates the practical use of tuples, lists, loops, conditions, and arithmetic operations in library management.